



45-DAYS SUMMER INTERNSHIP ON **NODE JS BACKEND DEVELOPMENT**

MODULE : 01 **JAVASCRIPT & BACKEND FUNDAMENTALS**

- JavaScript (ES6+) recap (variables, functions, async concepts)
 - Introduction to Backend & Node.js
 - Node.js architecture & modules
 - File system & basic server creation
 - Mini Task: Simple Node.js server
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MODULE : 02 **EXPRESS.JS & API DEVELOPMENT**

- Introduction to Express.js
 - Routing & middleware
 - REST API concepts
 - CRUD operations
 - Mini Project: Basic REST API
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MODULE : 03 **DATABASE INTEGRATION**

- Introduction to databases (MongoDB)
- Mongoose (schema, models)
- Connecting Node.js with database
- CRUD with database
- Mini Project: API with database





MODULE : 04 **AUTHENTICATION & ADVANCED CONCEPTS**

- User authentication (JWT)
 - Password hashing (bcrypt)
 - Authorization & role management
 - Error handling & validation
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MODULE : 05 **DEPLOYMENT & REAL-WORLD PROJECTS**

- API testing (Postman)
 - Deployment (Render / Railway)
 - **Final Projects:**
 - **Login & Register System**
 - **E-Commerce API**
 - **Blog API**
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AI TRAINING MODULE : **GENERATIVE AI & AI TOOLS FOR DEVELOPERS**

- Introduction to Generative AI and its role in modern development
 - Introduction to AI tools: ChatGPT, Google Gemini, Claude, GitHub Copilot, Perplexity, ElevenLabs, SuperBot
 - Prompt engineering and effective AI usage
 - Code generation using AI tools for frontend and backend development
 - Building a mini project using AI-assisted workflow
 - Understanding limitations, ethics, and best practices of AI
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OUTCOME

By the end of this internship, students will be able to build scalable REST APIs, work efficiently with databases and implement secure authentication systems. They will gain a clear understanding of backend architecture and develop real-world backend projects, preparing them for industry-level development.

